

Supplementary Summer Examination – 2023

Course: B. Tech.

Branch : Mechanical

Semester :V

Subject Code & Name: BTMC502 – Machine Design - I

Max Marks: 60

Date:09/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.	12
A) Describe Machine Design Procedure.	6
B) Explain Basic type of product form in Aesthetics design	6
C) Explain with sketch man-machine closed loop system.	6
Q.2 Solve Any Two of the following.	12
A) A cylindrical shaft made of steel of yield strength 700mpa is subject to static load consisting of B.M.10 KN-m and a tortional moment of 30 KN-m. Determine the diameter of shaft using maximum shear stress theory of failure. (F.S. = 2, E = 210 Gpa and Poission's ratio = 0.25)	6
B) Design Cotter joint. P = 30 KN Material = Carbon Steel Shear Stress = 35 N/mm ² Tensile Stress = 50 N/mm ² Compressive Stress = 90 N/mm ²	6
C) Explain design procedure of Knuckle joint.	6
Q. 3 Solve Any Two of the following.	12
A) Define any two- Stress Concentration and its remedies. Fatigue failure and Endurance limit Reversed Stress and Repeated Stress	6
B) Explain – i) Goodman dia ii) Solderberg dia	6
C) A steel rod subjected to a revered axial load of 180KN. Find the diameter of the rod. , If F.S= 2 $\sigma_u = 1070\text{MPa}$ $\sigma_y = 910\text{ MPa}$ $\sigma_e = \frac{1}{2} \sigma_u$	6
Factors- $K_a = 0.8$ $K_b = 0.85$ $K_c = 0.7$, $K_f = 1$	

- Q.4 Solve Any Two of the following.** **12**
- A)** Prove that for square key crushing stress is twice the shear stress. **6**
- B)** Design muff coupling which is used to connect two shaft transmitting 40KW at 350rpm **6**
- Material for Shaft and Key = Plain Carbon Steel
 τ = Shear Stress = 40 Mpa
 σ_c = Crushing Stress = 80 Mpa
 Material for Muff is C.I
 τ = Shear Stress = 15 Mpa
- C)** Explain design procedure of Bushed-Pin type flexible coupling. **6**
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- Q. 5 Solve Any Two of the following.** **12**
- A)** Explain Bolt of uniform strength **6**
- B)** A plate of 100mm wide and 10mm thick is to be welded to another plate by means of double parallel fillet weld. The plates are subjected to a load of 80 KN. Find the length of weld if shear stress = 55 Mpa. **6**
- C)** Design a helical compression spring for a maximum load of 1000 N for a deflection of 25mm. **6**
- C = Spring Index = 5
 Shear Stress = 420 N/mm²
 G = 84 KN/ mm²
 Consider Wahl's Factor

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